

Lionel SOPGOU

Postdoctoral Researcher in Mathematical & Actuarial Finance

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RESEARCH SUMMARY

I am a Postdoctoral Researcher at ENSAE Paris and the Institut Louis Bachelier, working with *Prof. Caroline Hillairet* and *Prof. Olivier Lopez* on the modeling and quantification of cyber risks for finance and insurance. My research draws on epidemiological contagion models, stochastic differential games, Hawkes processes, and extreme value theory. I completed my Ph.D. at Université Paris Cité and Imperial College London in November 2024, focusing on climate transition risk in credit portfolios under a CIFRE agreement with BPCE S.A. I previously earned an engineering degree and an MSc from Institut Polytechnique de Paris, where my thesis explored the use of Generative Adversarial Networks for financial applications.

CURRENT POSITION

Postdoctoral Researcher in Mathematical & Actuarial Finance — *ENSAE Paris & Institut Louis Bachelier* May 2025 – present

Paris, France

Supervisors: Prof. [Caroline Hillairet](#) & Prof. [Olivier Lopez](#).

- Stochastic multi-group SIR model for quantifying cyber-attack financial impact on insurers.
- Cyberattack arrival modeling via Cox processes and Bernoulli contagion from subsidiary firms.
- Major-minor MFG with common jumps and impulse control for optimal cybersecurity investment.

EDUCATION

PhD in Mathematical Finance — *Université Paris Cité / Imperial College London*

2021 – 2024

Paris, France & London, United Kingdom

Thesis: Pricing and Hedging of Climate Transition Risk in Credit Portfolio. [[HAL](#)]

Advisors: Prof. [Jean-François Chassagneux](#) (LPSM, Université Paris Cité) & Prof. [Antoine Jacquier](#) (Imperial College London).

Industry partner: BPCE S.A. (CIFRE), supervised by Dr. Smail Ibbou.

M.S. in Financial Mathematics: Statistics and Finance — *École Polytechnique / ENSAE Paris*

2019 – 2020

Palaiseau, France

Thesis: Machine Learning for Finance — prediction of implied volatility surfaces via Deep Learning; generation via Generative Adversarial Networks.

Courses: Interest rate models, Algorithmic trading, Advanced ML, Portfolio modelling, Financial econometrics, GARCH & stochastic volatility models.

Engineering Degree in Statistical Modelling and Applications — *Télécom SudParis*

2017 – 2020

Évry, France

Courses: Probability & statistics, Optimisation, Data mining, Stochastic processes, Economic modelling, Data networks, Web architectures.

Preparatory Classes for Grandes Écoles (MPSI/MP) — *Lycée Jeanne d'Albret & Lycée Jacques Amyot*

2015 – 2017

Saint-Germain-en-Laye & Melun, France

Yaoundé, Cameroon

Courses: Linear algebra, Real analysis, Probability, Numerical analysis, Mechanics, Database, Object-oriented programming.

PUBLICATIONS

Journal Articles

C. Hillairet, O. Lopez & L. Sogoui. “A stochastic SIR model for cyber contagion: application to granular growth of firms and to insurance portfolio.” *Scandinavian Actuarial Journal*, 1–34, Taylor & Francis, 2026. [doi]

L. Sogoui. “Impact of the carbon price on credit portfolio’s loss with stochastic collateral.” *Quantitative Finance*, 1–30. Taylor & Francis, 2025. [doi]

G. Bouveret, J.-F. Chassagneux, S. Ibbou, A. Jacquier & L. Sogoui. “Propagation of a carbon price in a credit portfolio through macroeconomic factors.” *SIAM Journal on Financial Mathematics*, 16(2):545–605, 2025. [doi]

Preprints

L. Sogoui. “Modeling the impact of climate transition on real estate prices.” *arXiv:2408.02339* / SSRN:4913777. Submitted to *Journal of Climate Finance*, 2024. [arXiv]

L. Sogoui. “Climate-Vulnerable Firms, Credit Supply, and Cascading Failures: An Agent-Based Model.” HAL:05645218. Submitted to *Economic Modelling*, 2026. [HAL]

Thesis & Other

L. Sogoui. *Pricing and Hedging of Climate Transition Risk in Credit Portfolio*. PhD thesis, Université Paris Cité, 2024. [HAL]

L. Sogoui. *Les essais de Paukémil, l'intrus universel*. Books on Demand, 2021. (Political philosophy.)

ONGOING WORKS

With **J.-F. Chassagneux.** “Real Estate Pricing under Transition Risk: A Real Option Approach.” Model computing the optimal discount between energy-rating classes, accounting for optimal renovation planning and idealized deterministic transition scenarios.

With **D.Bastide & S.Crepey & S.Pavarana & R.Timeus.** “Dynamic Top-Down Modeling of Financial Losses under Climate, Credit, and Market Risks.” We propose an integrated risk-management framework that links financial, economic, and climate-related dynamics to model loss distributions and assess key regulatory metrics across interconnected credit, market, liquidity, and climate-transition channels.

With **C. Hillairet.** “A major-minor MFG with common jumps and impulse control for optimal cybersecurity investment.” We model firms versus ransomware interactions as a major-minor mean fields with McKean-Vlasov dynamics, common jumps, and impulse control to capture the strategic interplay between a rational attacker and a population of heterogeneous firms.

TALKS, SEMINARS & CONFERENCES

Upcoming 2026

- **Mathematical Advances on Emerging Risks 2026**, Mexico City, Mexico, Nov. 2026.

2026

- **7th European Actuarial Journal Conference**, Istanbul, Turkey, Sep. 2026.
- **CEMRACS 2026**, Marseille, France, Jui. 2026.
- **XIII Bachelier World Congress**, Bologna, Italy, Jun.–Jul. 2026.
- **Scandinavian Actuarial Conference 2026**, Stockholm, Sweden, Jun. 2026.

- **Financial Risks International Forum**, Paris, Mar. 2026.
- **XXVII Workshop on Quantitative Finance**, Bergamo, Italy, Mar. 2026.

2025

- **Quantitative Methods in Finance (QMF 2025)**, Sydney, Australia, Nov. 2025.
- **Séminaire Actuariat & Finance** (IRA-ISFA-ENSAE-CNAM-ISUP), Le Mans, Nov. 2025.
- **SIAM Conference on Financial Mathematics & Engineering (FM25)**, Miami, FL, Jul. 2025.
- **EconophysiX Seminar**, Capital Fund Management, Paris, Apr. 2025.
- **UCLA Financial & Actuarial Mathematics Seminar** (online), Feb. 2025.
- **London–Oxford–Warwick Financial Mathematics Workshop**, Oxford, Jan. 2025.

2024

- **9th Green Finance Research Advances**, Banque de France, Paris, Dec. 2024.
- **Groupe de Travail – Risques Climatiques**, CACIB, Montrouge, Oct. 2024.
- **12th Bachelier World Congress**, FGV EMap, Rio de Janeiro, Jul. 2024.
- **XXV Workshop on Quantitative Finance**, Università di Bologna, Apr. 2024.
- **Séminaire Bachelier**, Institut Henri Poincaré, Paris, Feb. 2024.

2023

- **8th Green Finance Research Advances**, Banque de France, Paris, Dec. 2023.
- **European Summer School in Financial Mathematics**, TU Delft, Sep. 2023.
- **London/Oxford/Warwick Financial Mathematics Workshop**, King's College London, Jul. 2023.
- **Groupe de travail des thésards du LPSM**, Sorbonne Université, Paris, May 2023.
- **Quantitative Finance Workshop 2023**, Università di Cassino, Gaeta, Mar. 2023.

2022

- **London-Paris Bachelier Workshop (6th edition)**, Institut Henri Poincaré, Paris, Sep. 2022.

TEACHING ASSISTANTSHIP

Introduction to Mathematical Finance (12h/year) — ENSAE Paris — Master 1

Feb.–May 2023 & 2024

Palaiseau, France

Topics: Financial derivatives, Valuation in financial markets, Pricing by trees, Stochastic calculus, Black–Scholes model.

WORK EXPERIENCE

CEO & Quantitative Trader — Bonetti Arbitrage Lab

May. 2025 – present

Paris, France

- Development and live execution of intraday statistical arbitrage strategies on US equities (IBKR API).
- Semi-systematic Python pipeline yielding triple-digit annual performance and Sharpe ratio > 4 .

Quantitative Risk Researcher (CIFRE PhD) — Groupe BPCE S.A.

Nov. 2021 – Aug. 2024

Paris, France

Advisor: Dr. Smail Ibbou (Head of Credit Risk Model Validation).

- Projection of portfolio risk measures over long time horizons under climate transition scenarios.

- Incorporation of greenhouse gas costs into economic modelling, firm valuation, real estate pricing, and credit risk.
- Development of macro-financial models linking carbon price dynamics to credit portfolio losses.

Quantitative Researcher Intern — LexiFi

May – Nov. 2020

Boulogne-Billancourt, France

Advisor: Olivier Pradère (Head of Quantitative Research).

- Deep Learning prediction and reconstruction of implied volatility surfaces from historical spot/forward series (illiquid, proprietary and OTC options).
- Development of *CRS-GAN*, a novel Generative Adversarial Network for synthetic implied volatility surface generation, used for options pricing and portfolio scenario analysis.

Summer Intern — Machine Learning — BlackRock

Jun. – Aug. 2019

Budapest, Hungary

Advisor: Peter Koczka (Associate).

- Study of NLP and ML classification methods applied to financial documents.
- Built a robust financial document classifier (> 80% accuracy) trained on a corpus of over 4 000 labeled items.

Software Developer Intern — Robert Bosch France

Jun. – Sep. 2018

Drancy, France

Advisor: Stephan Nzokou (Engineer).

- Development, testing, and validation of the *PDC FLASH* tool for reprogramming the PSA CallBlock braking system firmware.
- Technical documentation and team training on the tool.

Volunteer — Planète-Urgence / Association Tubè Awù

Aug. – Sep. 2019

Ebodjé, Cameroon

- Learned about the association's marine conservation work, especially on marine turtles along the Central African coast.
- Developed the association's website, Facebook page, and YouTube channel to strengthen its public visibility and outreach.

SKILLS

Programming

Python (advanced), C++ (advanced), VB.NET (advanced), Java / J2EE, MATLAB, R, MySQL, SAS.

Libraries

TensorFlow 2, Keras, PyTorch, Scikit-learn, NumPy, Pandas, LexiFi Apopos, IBKR API, etc.

Tools

Git, Linux, R Studio, Visual Studio, Eclipse, L^AT_EX.

Languages

French (native), English (advanced), Russian (basic).

Key strengths

Autonomous, proactive problem solver, detail-oriented, intellectually curious, efficient under pressure, team player.

REFERENCES

Prof. Jean-François Chassagneux

PhD Supervisor
LPSM, Université Paris Cité
chassagneux@lpsm.paris

Prof. Thorsten Schmidt

PhD thesis reviewer
University Freiburg
thorsten.schmidt@stochastik.uni-freiburg.de

Prof. Caroline Hillairet

Postdoc Supervisor
ENSAE Paris
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